Transverse & Longitudinal Waves (Cambridge (CIE) A Level Physics): Revision Note

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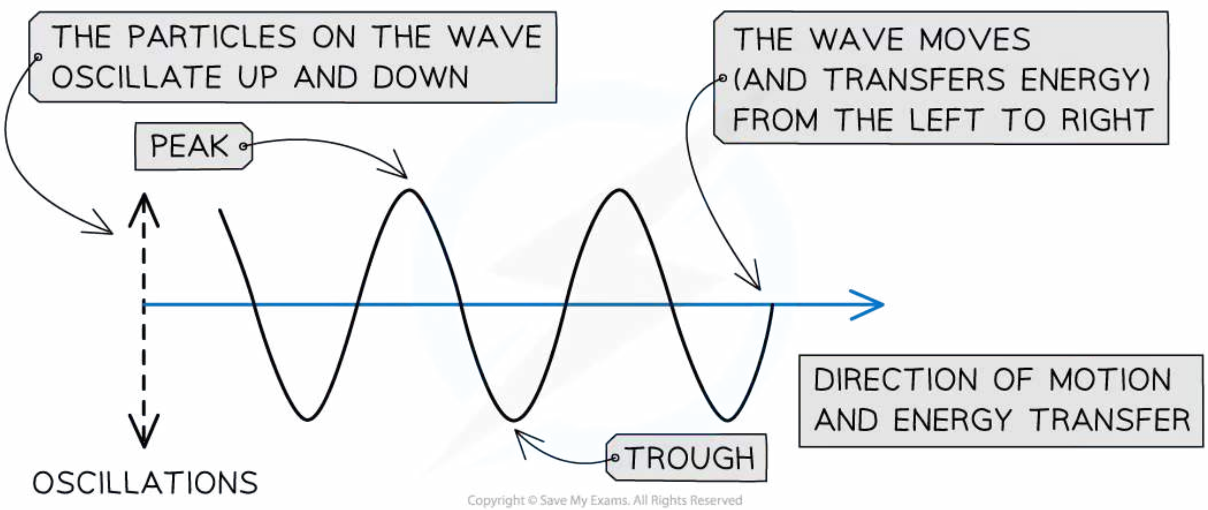
Properties of transverse & longitudinal waves

- In mechanical waves, particles oscillate about fixed points
- There are two types of mechanical wave:
  - transverse
  - longitudinal
- The type of wave can be determined by the direction of the oscillations in relation to the direction the wave is travelling

Transverse waves

- Transverse waves are defined as follows:
  A wave in which the particles oscillate perpendicular to the direction of motion and energy transfer



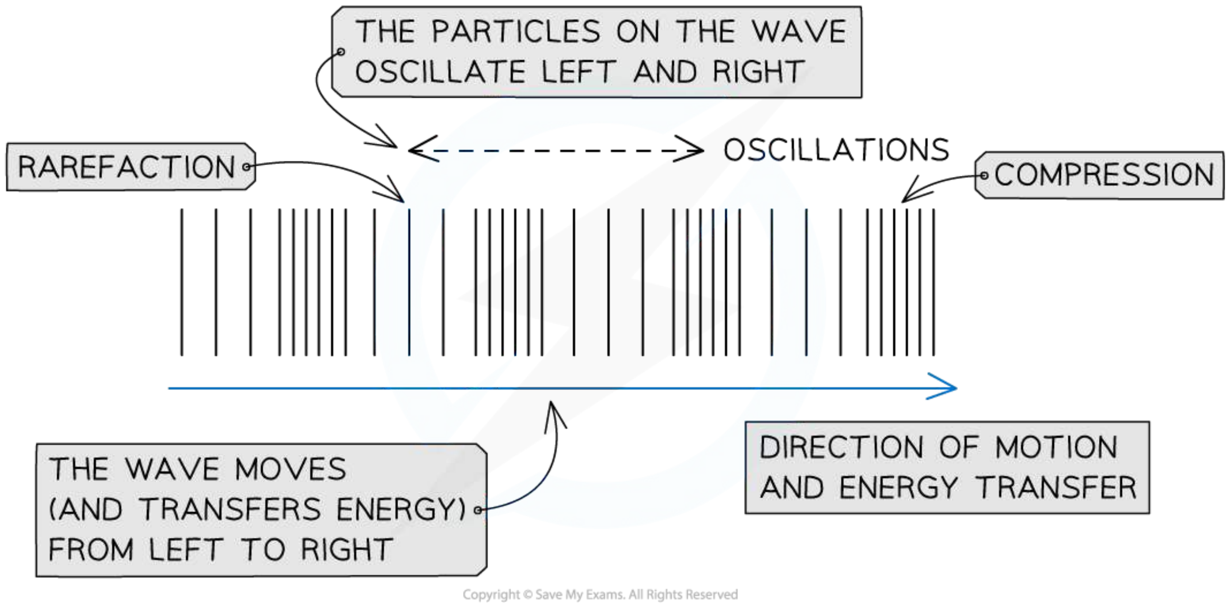
A transverse wave travelling from left to right

- Transverse waves show areas of **peaks** and **troughs**
- Examples of transverse waves include:
  - Electromagnetic waves e.g. radio, visible light, UV
  - Vibrations on a guitar string

- Transverse waves do not need particles to propagate, and so they **can travel through a vacuum**
- Transverse waves **can** be polarised

Longitudinal waves

- Longitudinal waves are defined as follows:
  A wave in which the particles oscillate parallel to the direction of motion and energy transfer



A longitudinal wave travelling from left to right

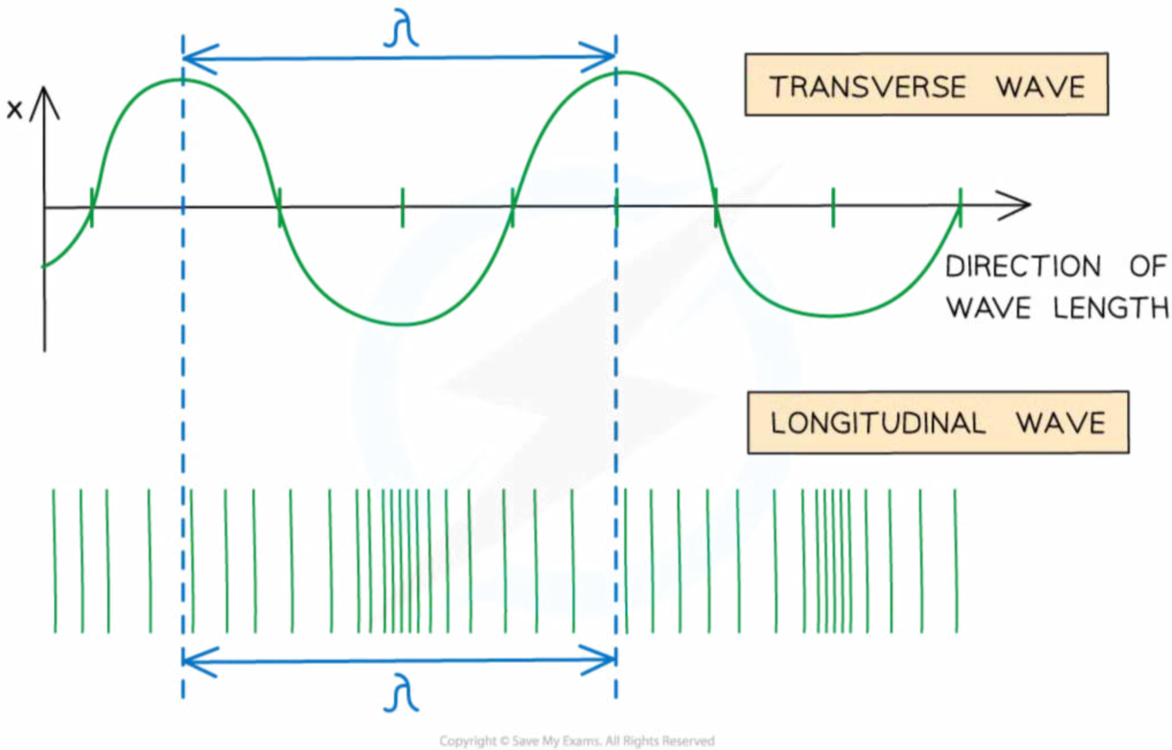
- As a longitudinal wave propagates, areas of low and high pressure can be observed:
  - A **rarefaction** is an area of low pressure, with the particles being further apart from each other
  - A **compression** is an area of high pressure, with the particles being closer to each other

- **Sound waves** are an example of longitudinal waves

- Longitudinal waves need particles to propagate, and so they **cannot travel through a vacuum**
- Longitudinal waves **cannot** be polarised

- The diagram below shows the equivalent of a wavelength on a longitudinal wave

Comparing wavelengths of longitudinal and transverse waves



Wavelength shown on a longitudinal wave

Examiner Tips and Tricks

The definitions of transverse and longitudinal waves are often asked as exam questions, make sure to remember these by heart!

The properties of waves you learned about in General Wave Properties, such as amplitude and wavelength, all apply to transverse and longitudinal waves

Graphical representations of transverse & longitudinal waves

- Transverse and longitudinal waves can be represented graphically in the same way
- The main difference is the meaning of 'displacement' on the y-axis
  - Alternatively, displacement could be exchanged for velocity or acceleration

- For a transverse wave:
  - Displacement is **perpendicular** to the direction of propagation (energy transfer)

- For a longitudinal wave:
  - Displacement is **parallel** to the direction of propagation (energy transfer)

Graphical representation of a progressive wave

The wave has a period of 1.0 s, a frequency of 1.0 Hz and could be either transverse or longitudinal.

Worked Example

The graph shows how the displacement of a particle in a wave varies with time.

Which statement is correct?

A. The wave has an amplitude of 2 cm and could be either transverse or longitudinal.

B. The wave has an amplitude of 2 cm and a period of 6 s.

C. The wave has an amplitude of 4 cm and a period of 4 s.

D. The wave has an amplitude of 4 cm and must be transverse.

Answer: A

Step 1: Determine amplitude from the graph:

- Each peak and trough is 2 cm away from the equilibrium position on the graph
- So, the amplitude is 2 cm

Step 2: Determine the time period:

- 4 seconds pass before the wave repeats its pattern again, so the period is 4 s
- The amplitude of 2 cm and period of 4 s rules out options **B**, **C** and **D**

Step 3: Discuss whether the wave is transverse or longitudinal:

- The x-axis of this graph is time, so all that can be inferred is that the particle is displaced periodically
- This occurs if a transverse wave or a longitudinal wave is causing its oscillations, as it is not mentioned how it is oscillating relative to the direction of wave travel
- Therefore the wave could be **either** transverse or longitudinal

Examiner Tips and Tricks

Both transverse and longitudinal waves can look like transverse waves when plotted on a graph - make sure you read the question and look for whether the wave travels **parallel** (longitudinal) or **perpendicular** (transverse) to the direction of travel to confirm which type of wave it is.